Design

AI-Based Phishing and Social Engineering Detection Assistant (PS57)

Companion to `PRD_PhishGuard_PS57.md`, `ARCHITECTURE_PhishGuard_PS57.md`, `RULES_PhishGuard_PS57.md`, and `PHASES_PhishGuard_PS57.md`. This is the visual/UX spec for the Phase 5 dashboard build (Aug 27–28) and for the PPT's visual language — the two should feel like the same product, not two unrelated projects.

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1. Design Principles

- **Clarity over cleverness.** A judge or a non-technical user should understand the verdict in under 3 seconds of looking at the screen — that's literally the product's own success metric (PRD §3), so the UI needs to embody it.
- **Explainability is the differentiator, so make it visible.** Don't bury the "why" in a tooltip or a collapsed section — the plain-language explanation is the core innovation (PRD §2), it should be one of the most prominent elements on screen, not an afterthought below the fold.
- **Calm, not alarmist.** Security tools often lean on red/siren aesthetics. A cleaner, more clinical palette signals trustworthiness and control — reads better to judges evaluating "feasibility" than a panic-mode UI.
- **Demo-safe by design.** Every visual state (loading, error, fallback) needs to look intentional, not broken — because Rules §7 says the network may fail live and the UI still needs to hold up.

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2. Color System

Purpose	Color	Notes
Base background	Neutral off-white / near-black (pick one theme, don't mix)	Keeps focus on the gauge and verdict, not the chrome
Primary brand accent	A single confident color (e.g. deep blue or teal)	Used for buttons, active states, headers

Purpose	Color	Notes
Low risk (0–33)	Green	Gauge + badge color
Medium risk (34–66)	Amber/yellow	Gauge + badge color
High risk (67–100)	Red	Gauge + badge color — use deliberately, not everywhere
Text	High-contrast neutral (near-black on light theme, near-white on dark)	Accessibility matters if judges view on a projector

Keep the palette to 1 accent color + the 3 risk colors + neutrals. Resist adding more — a scattered palette reads as unfinished, and a tight one reads as intentional.

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3. Typography

- One typeface family, two weights max (regular + bold/semibold). Don't mix multiple fonts.
- Clear size hierarchy: large for the risk score number itself (this is the single most important number on screen), medium for verdict label, smaller for the explanation paragraph and threat-factor badges.
- The explanation text should read like a sentence a human wrote, not a log line — match the LLM prompt's tone (PRD §6.2) with a readable, generous line-height in the UI.

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4. Layout — Dashboard (single page)

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┌────────────────────────────────────────────┐
│ Header / Product name                  │
├────────────────────────────────────────────┤
│ [ Paste area: text / SMS / URL input ]     │
│ [ Sample Case 1 ] [ Sample Case 2 ] [ Case 3 ]│
│            [ Analyze button ]          │
```

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├──────────────────────────────────────────────┤
│   ┌──────────────────┐            │
│   │ 0-100 Gauge │ Verdict Badge  │
│   └──────────────────┘            │
│                         │
│ Threat Factor Badges: [Urgency] [Impersonation]│
│                         │
│ "This message pretends to be from State    │
│  Bank, but links to an unverified domain."  │
│ (plain-language explanation panel)       │
├──────────────────────────────────────────────┤
│ Optional: source breakdown (heuristic vs LLM) │
└──────────────────────────────────────────────┘
```

- Input area and sample-case buttons stay above the fold — this is the first interaction, and per Rules §7 the sample buttons are the primary demo path, not a fallback hidden in a menu.
- Gauge + verdict + explanation should be visible together without scrolling once a result loads — the "why" (PRD's core differentiator) shouldn't require extra clicks.

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5. Key Components

5.1 Threat Gauge

- Semi-circular or circular 0–100 gauge, color-coded per §2 (green/amber/red).
- Large numeric score in the center — this is the number people will glance at first.
- Animate the needle/fill on result load (subtle, not gimmicky) — makes the "under 3 seconds" feel fast rather than static.

5.2 Verdict Badge

- Short label: "Phishing" / "Social Engineering" / "Legitimate" / "Suspicious".
- Color matches the gauge tier, so the two never visually contradict each other.

5.3 Threat Factor Badges

- Small pill-shaped tags, one per detected factor (e.g. "High Urgency", "Mismatched Domain", "Impersonation of Bank").
- Directly sourced from the LLM's `threat_factors` array (Architecture §2.4) — don't invent extra ones in the UI layer.

5.4 Explanation Panel

- One or two sentences, plain language, directly from the LLM's `explanation` field.
- This is the component judges will actually read closely — give it visual weight, not a small caption font.

5.5 Sample Case Buttons

- Three clearly labeled buttons (not raw "Case 1/2/3" — label them meaningfully, e.g. "Fake Bank KYC", "HR Phishing Email", "Legitimate Notice").
- Clicking one should feel instant — these use hardcoded pre-computed responses per Rules §1/§7, so there's no excuse for a loading delay here.

5.6 Loading & Error States

- Loading: a short, calm skeleton/spinner state — not a blank screen (matters if a live-typed demo input is slow).
- Error/fallback: if the LLM call fails and the system falls back to heuristic-only scoring (Architecture §5), the UI should show a small, honest note like "AI analysis unavailable — showing heuristic result" rather than pretending nothing happened. Transparency here matches the product's own explainability principle.

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6. PPT Visual Language (Slide 3 architecture diagram especially)

- Reuse the same accent color and risk-tier colors from the dashboard in the architecture diagram (Slide 3) and comparison table (Slide 2) — visual consistency between the pitch and the product subtly signals that the team has already thought this through end-to-end.
- Architecture diagram should mirror the flow in `ARCHITECTURE_PhishGuard_PS57.md` §1: Input → Heuristics + LLM (parallel) → Scoring Engine → Explainable UI. Keep it to

boxes and arrows, no unnecessary 3D/skeuomorphic effects — clean and legible from the back of a room.
- Slide 2's "Existing Blocklists vs. Our Engine" comparison table: use the same green/red logic sparingly (red for reactive/opaque, green for proactive/explainable) so it echoes the product's own risk-color language.

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7. What to Avoid

- Don't over-design the demo UI at the cost of Phase 4/5 timeline — per Phases.md, Aug 27–28 is UI time, not a redesign sprint. A clean, simple layout beats an ambitious one you don't finish.
- Don't use stock "hacker" cybersecurity clichés (green matrix text, hoodie graphics, skull icons) — reads as generic rather than credible, and works against the "clinical, trustworthy" principle in §1.
- Don't let the gauge, badge, and explanation ever visually disagree (e.g. green gauge next to a "Phishing" badge) — if this happens it's a scoring bug, not a design choice, and should block the demo case until fixed.